LECTURE 3

# MACHINE LEARNING FOUNDATIONS FOR GEOSPATIAL PREDICTION

Geospatial Representation Learning

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# Learning Outcomes

## Lecture

- Explain supervised machine learning in terms of data, labels, models, losses, optimization, and generalization.
- Describe the traditional bias-variance trade-off and its relevance for geospatial prediction tasks.
- Explain how backpropagation and gradient-based optimization enable learning from data.
- Interpret neural networks as tensor transformation machines.
- Compare the basic roles of MLPs, CNNs, RNNs, and transformers in geospatial machine learning.
- Evaluate geospatial prediction models using suitable validation strategies, including spatially aware train-validation-test splits.

## Lab

- Implement a basic deep learning workflow for a geospatial prediction task in Python.
- Prepare training, validation, and test splits while considering spatial dependence.
- Train and evaluate a selected model using appropriate loss functions and metrics.
- Interpret prediction results and identify common sources of error or bias.

PRACTICAL 3

# TRAINING A GEOSPATIAL PREDICTION MODEL

Geospatial Representation Learning

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